

Computer Graphics 1

2D Graphics Pipeline

The infinite canvas

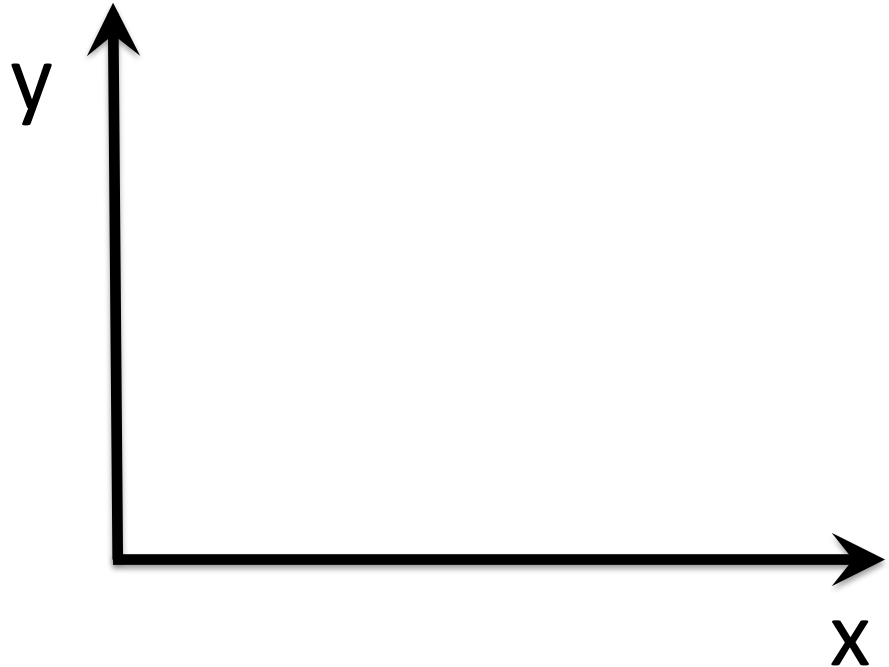
- Traditional artists work on a finite canvas
- In CG the canvas is infinite:
 - Extent
 - Zoom

<http://www.zoomify.com/>

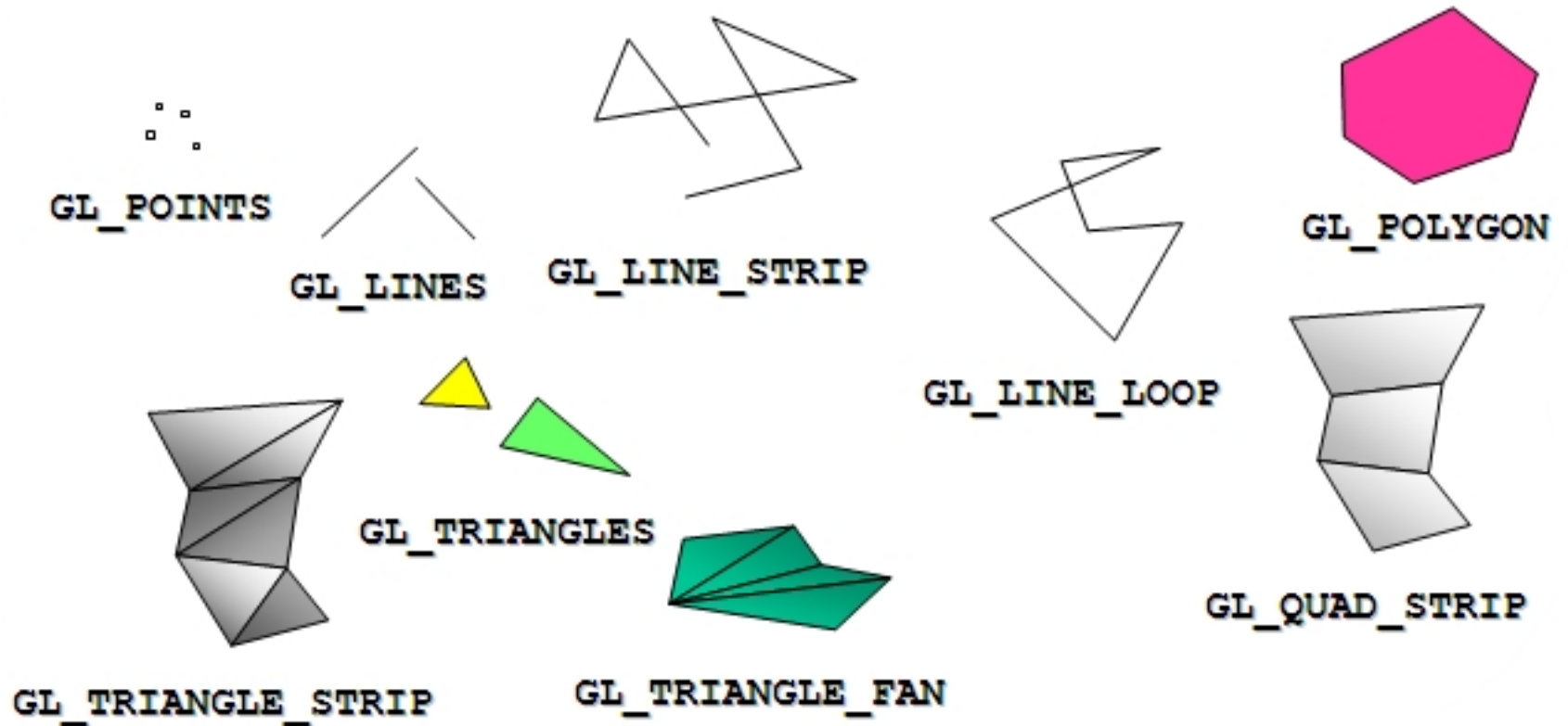
[http://www.ted.com/talks/
blaise_aguera_y_arcas_demos_photosynth.html](http://www.ted.com/talks/blaise_aguera_y_arcas_demos_photosynth.html)

The 2D world coordinate system

- Infinite:
 - Extent
 - Zoomability
- Floating point
- Continuous
- This is conceptually where you draw things



Drawing Things



The Vertex

- To specify each of these objects:
 - Supply a set of vertices:
 - Specification: finite set of points
 - Realization: continuous line, shape, or object

2D Pipeline

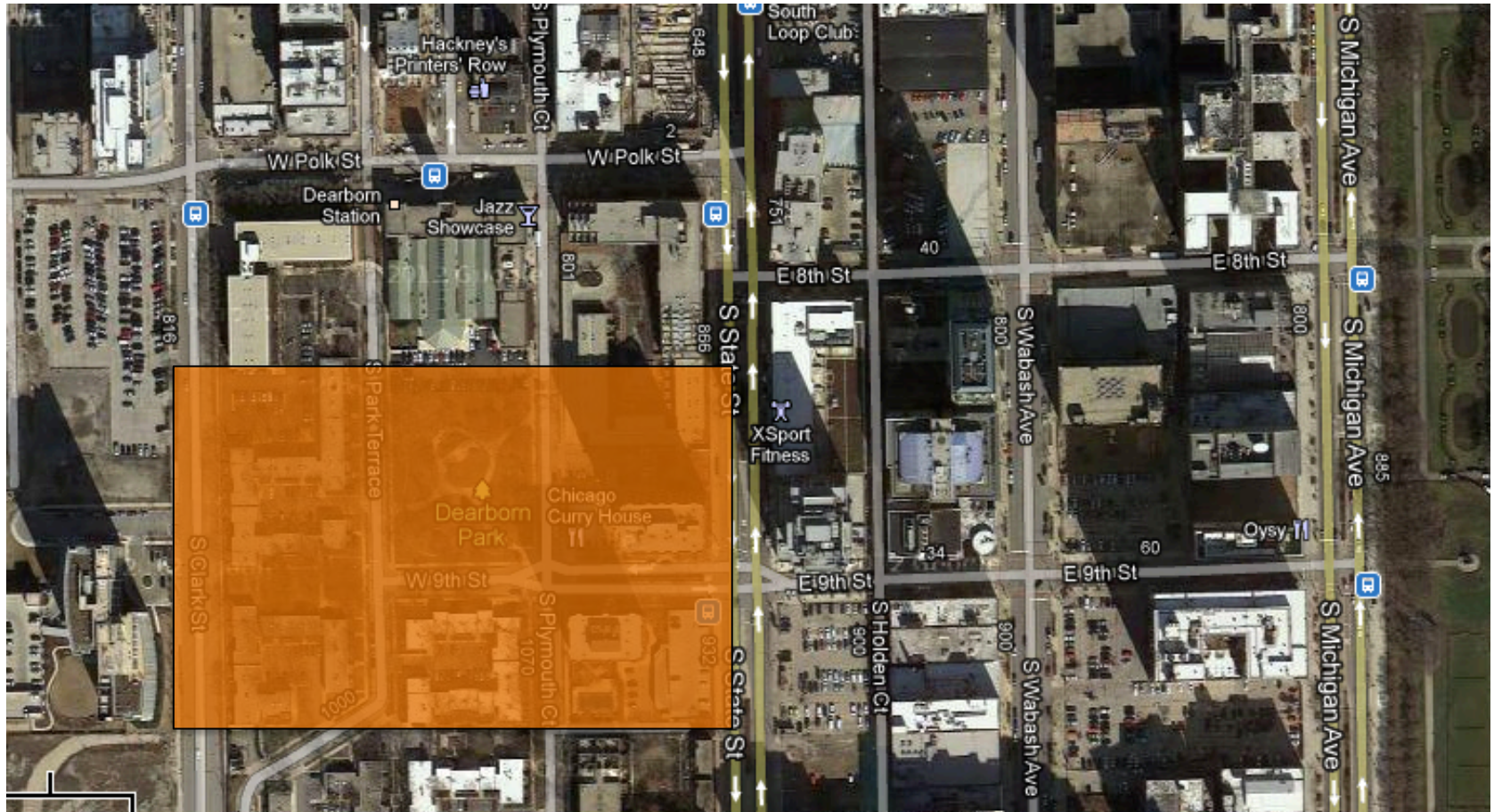
Drawing on
infinite canvas

vertices
(in world coords)



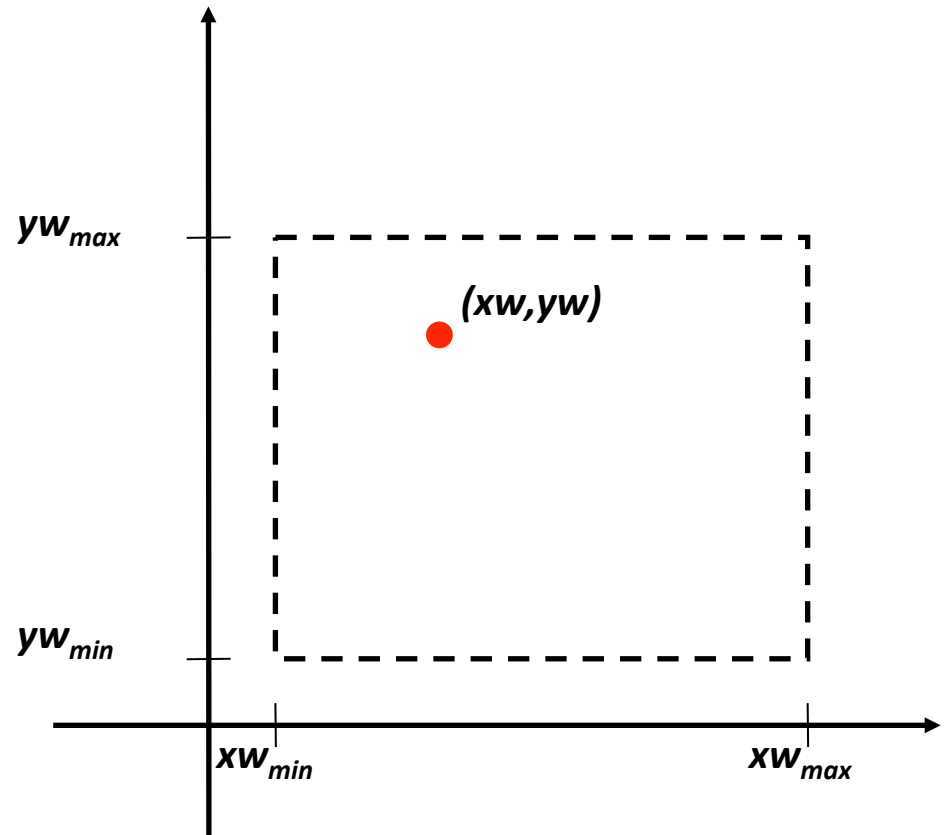
- The entire canvas (world) is not displayed.
- The part of the infinite world that will appear in the image is defined by a clip window

Clip Window



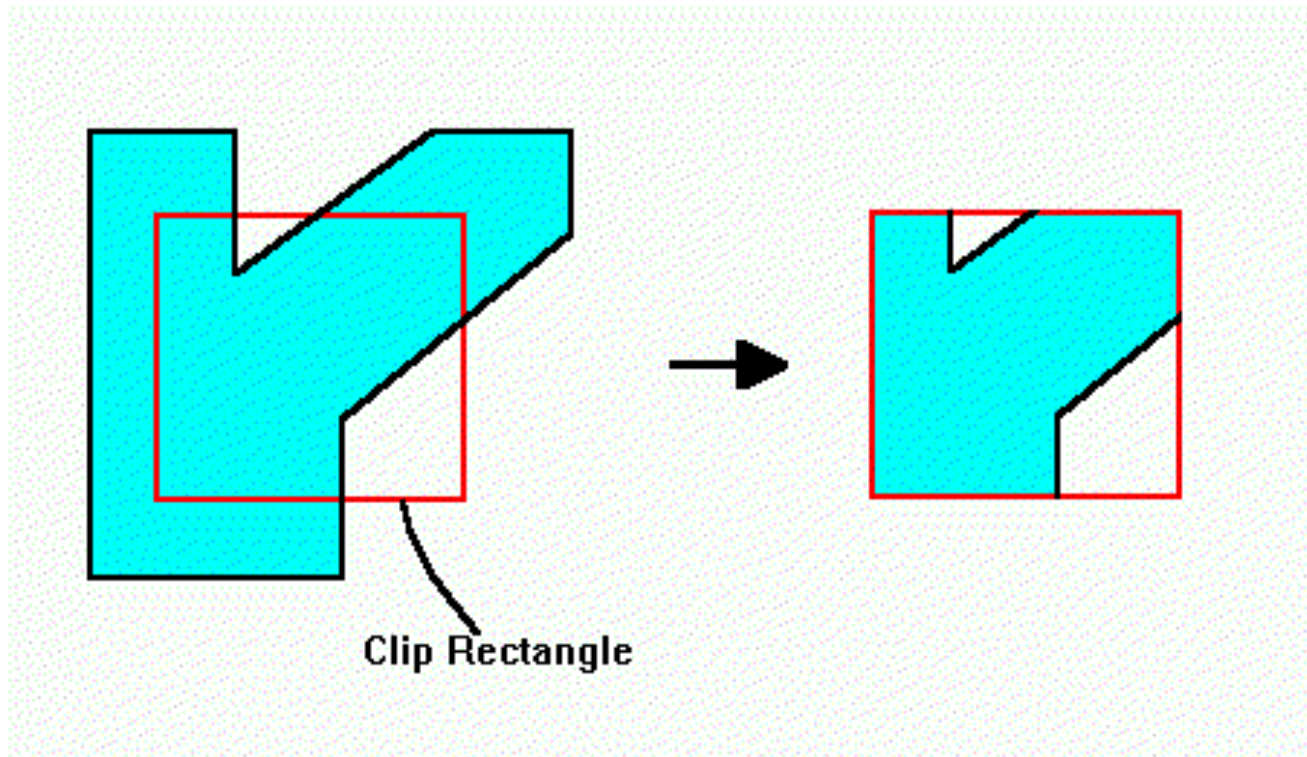
Clipped world coordinates

- Infinite:
 - Zoomability
- Finite:
 - Extent
 - Dimensions
- Floating point
- Continuous



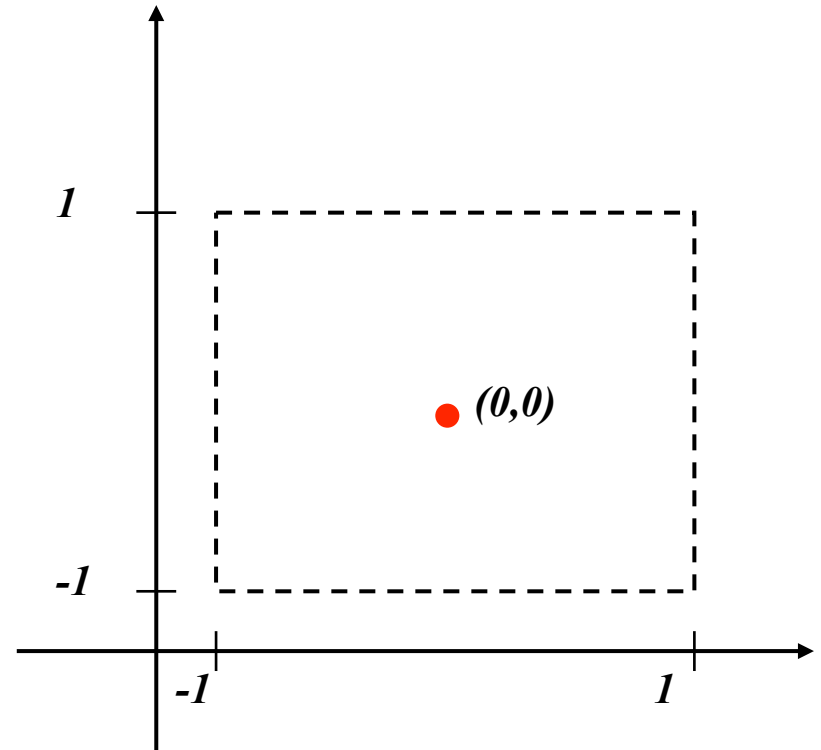
Clipping

- Only the objects (or portions of objects) inside the clip window are visible.

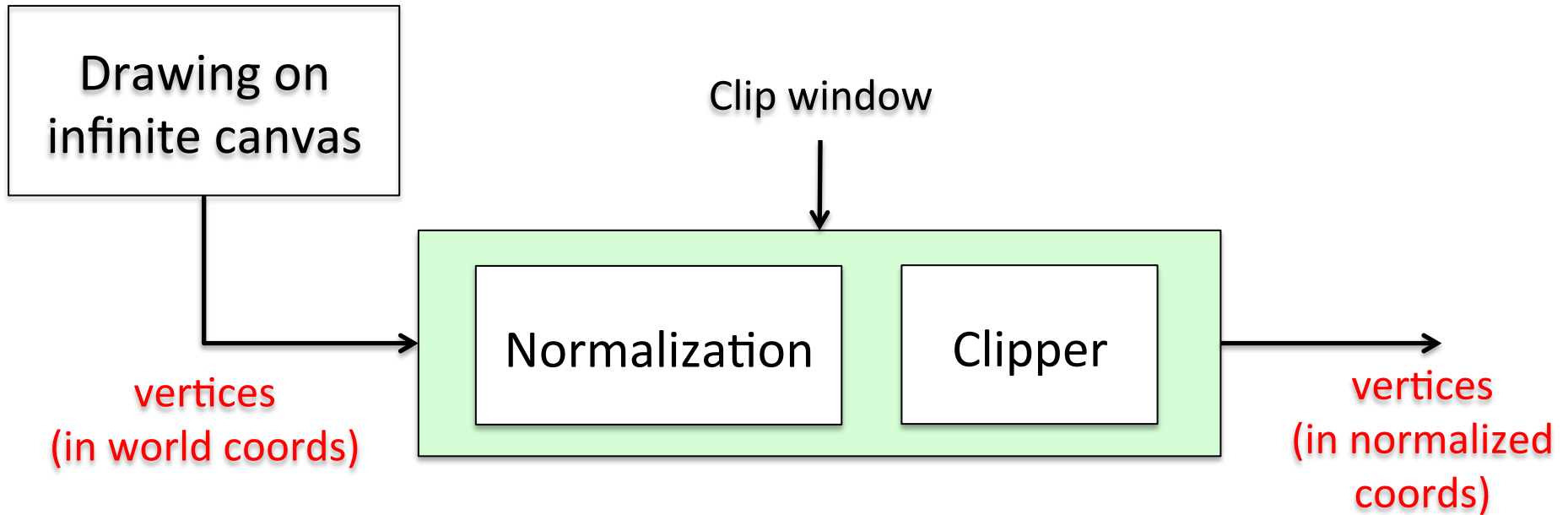


Normalized coordinates

- Infinite:
 - Zoomability
- Finite (standard):
 - Extent
 - Dimensions
- Floating point
- Continuous



2D Pipeline



The Digital World

- Digital Image
- Terms
 - Pixel
 - picture elements
 - Individual sample in a digital image
- Aspect ratio
 - Ratio of pixel length to pixel height
- Resolution
 - Dimensions of image in pixels

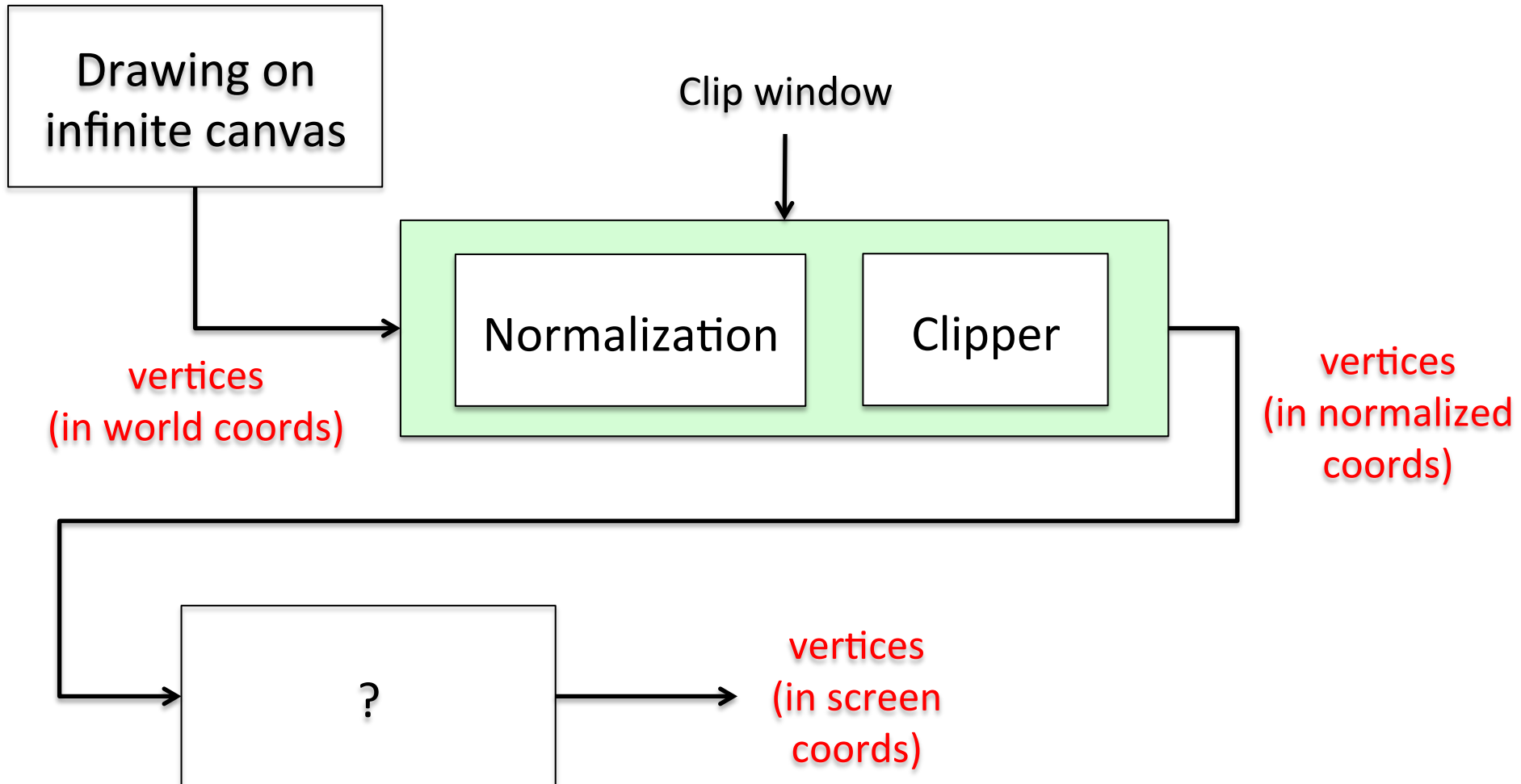
[illegible]

Device/Screen Coordinates

- Finite:
 - Zoomability
 - Extent
 - Dimensions
- Integral
- Discrete

[illegible]

2D Pipeline

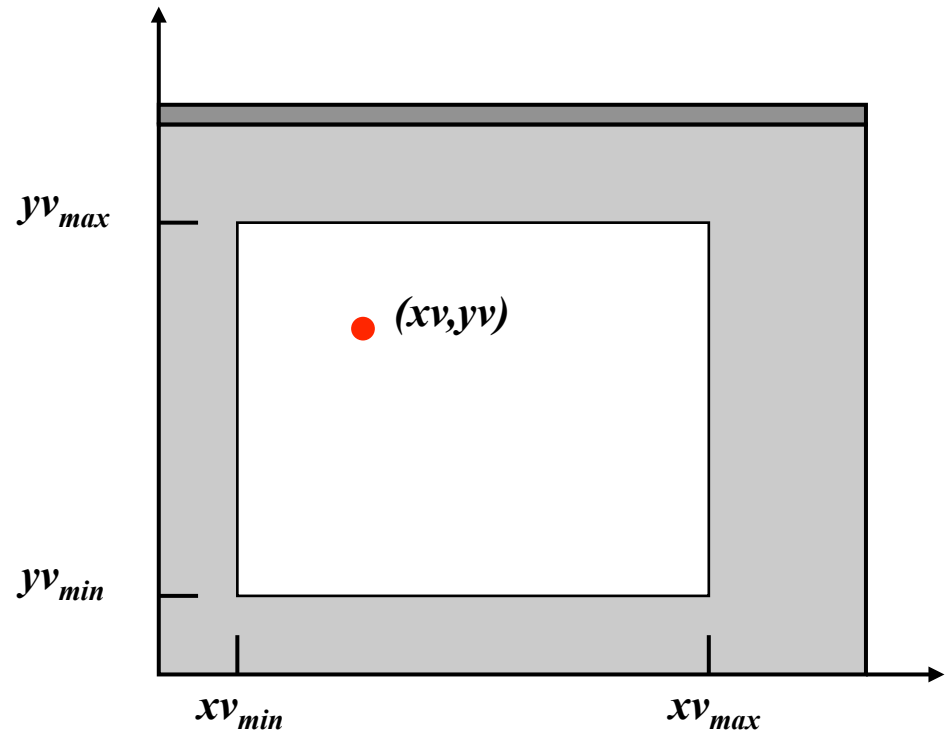
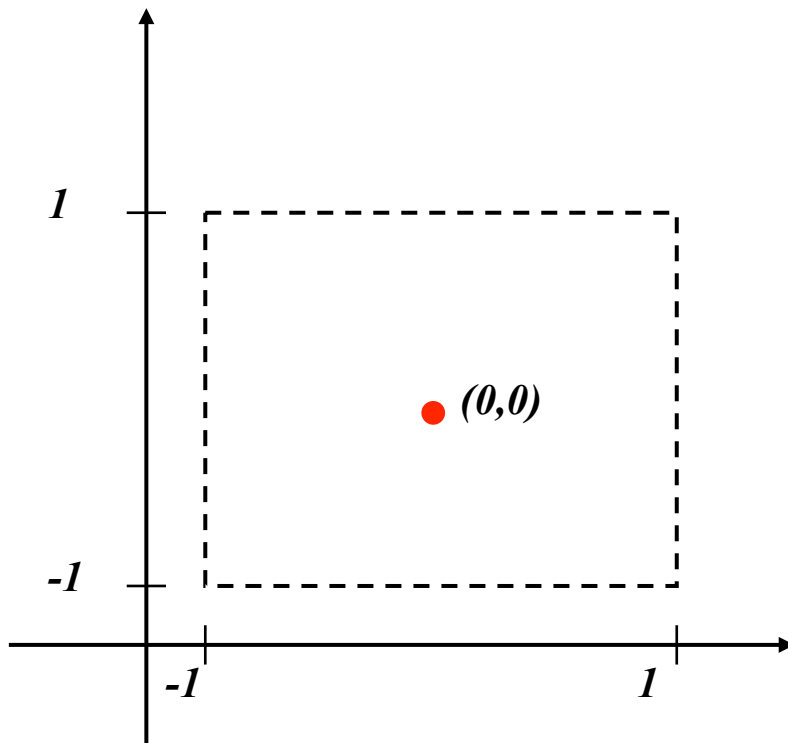


Viewport

- Trick is to map vertices in continuous world/clip space to discrete screen space.
- The **viewport** defines the size and position on the screen where the clipped image will be rendered.
 - Specified in integral screen coords

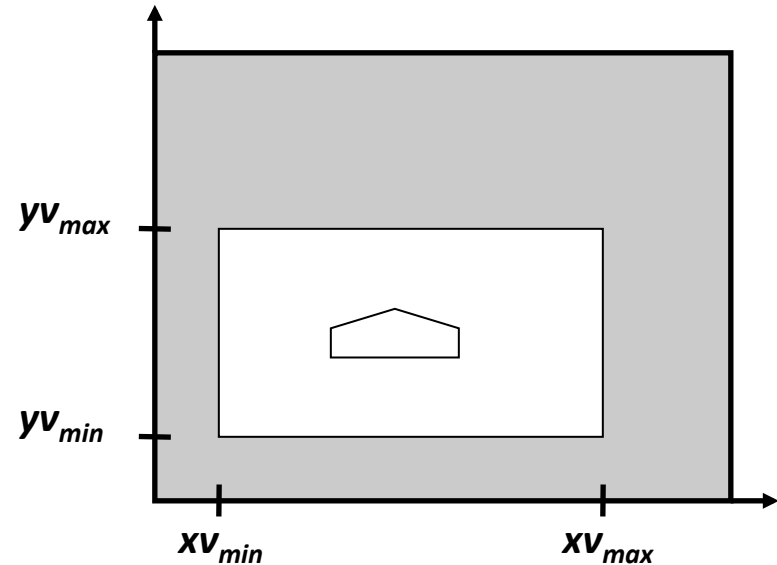
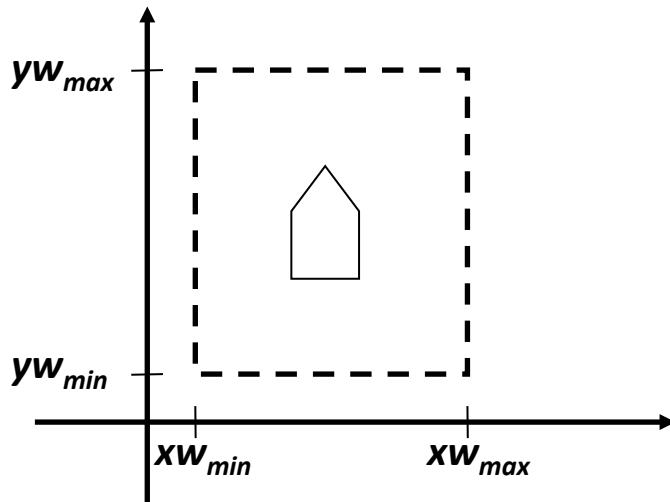
Viewing Transform

- World (clipping) window expressed in world coordinates
- Viewport expressed in screen coordinates
- Transformation from world to view is the viewing transform



Viewing Transform

- Viewing transformation is proportional

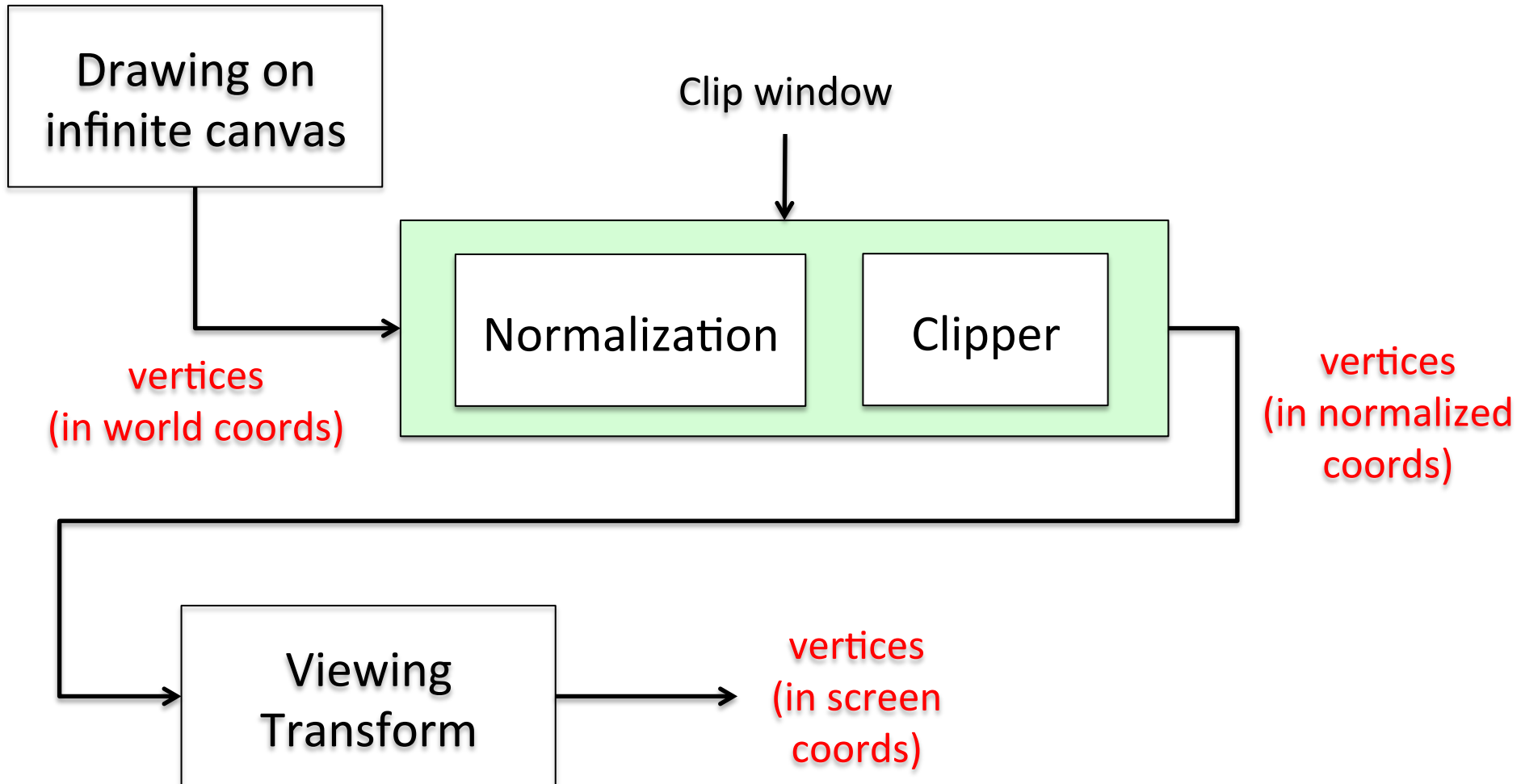


- Must map (xw, yw) in world/clip/normalized coords to (xv, yv) in viewport
 - Mapping must be properly scaled to get proportional relationship
 - E.g., if object is 55% of the way across the world/clipping/normalized window, must appear at 55% of the way across the viewport
 - Distortion here is due to different scaling in the two dimensions

Viewing Transform

- Interesting video about sizing content for different size screens:
 - <http://www.youtube.com/watch?v=vIFCV2spKtg>

2D Pipeline



Rasterization

Note that at this point we still only have the vertices that define more complex primitives, not the actual primitives (lines, polygons etc.)

Rasterization or **scan conversion** is the task of taking an image description and converting it into individual pixels:

- Connecting the dots to form a line
- Filling in the polygons

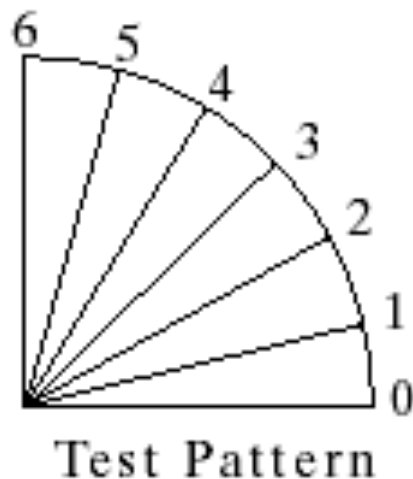
Note that this gets performed on vertex coordinates in screen space.

Rasterization

- Primitives are defined in continuous space
- Primitives are realized in discrete space.
- Process of going from continuous to discrete is called **sampling**.
- With sampling comes aliasing.

Aliasing

- Problem with sampling: the **jaggies**
 - Sometimes called **stairstepping**
 - Occurs when drawing lines that are not horizontal or vertical
 - Becomes apparent when display device has low resolution, or when the image is zoomed in on

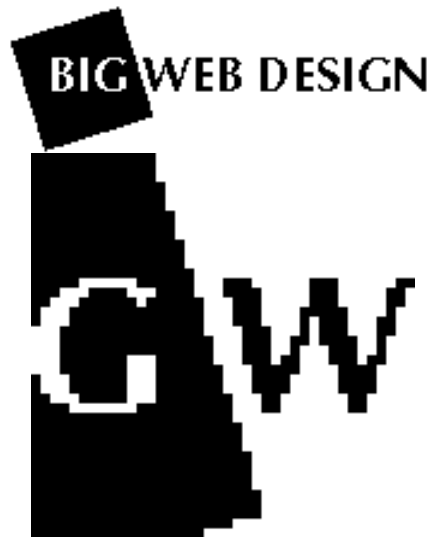


Test

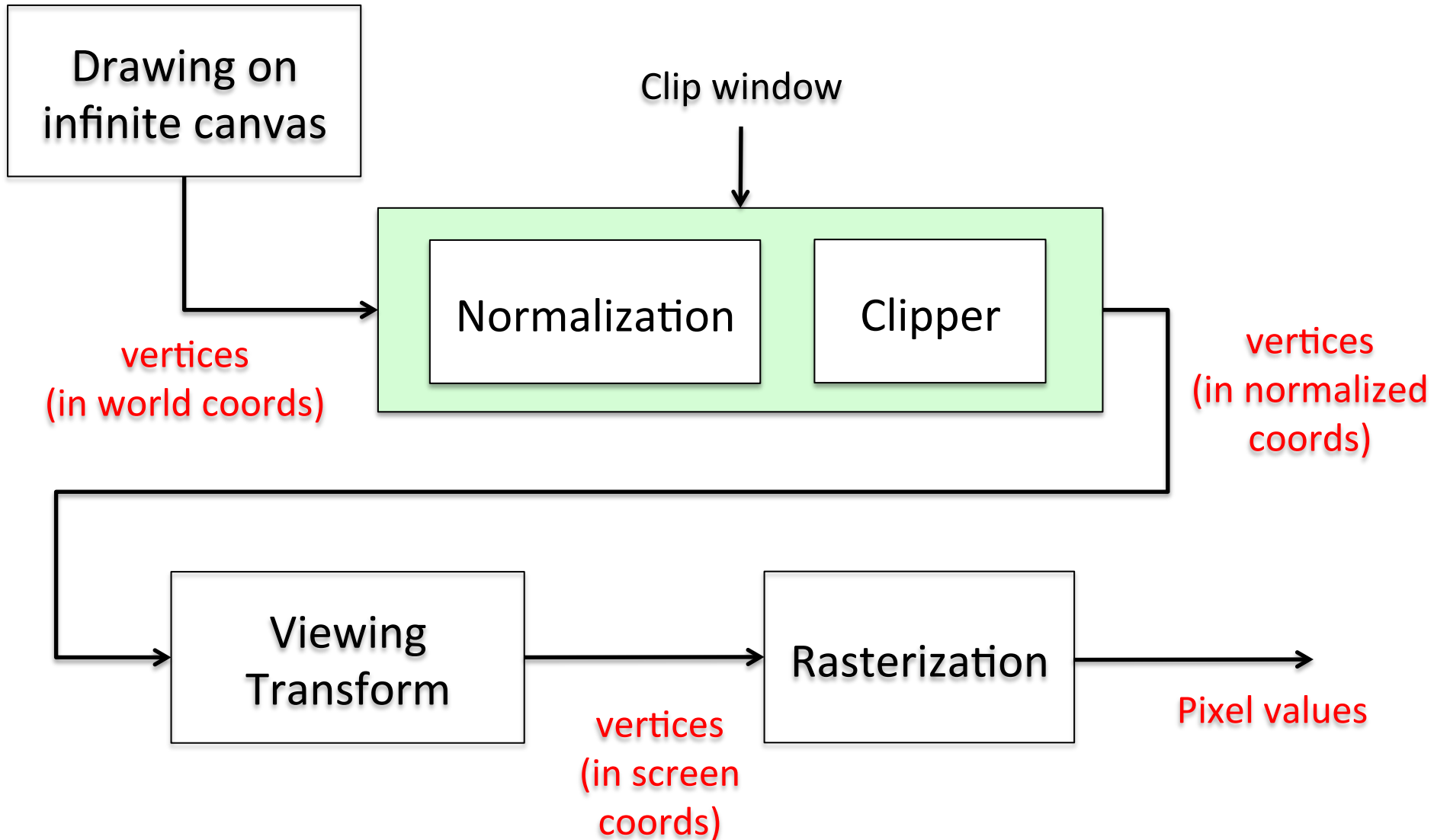


Antialiasing

- Can smooth out the image by padding pixels with intermediate colors



2D Pipeline



2D Pipeline Summary

- Specify primitives to be drawn by specifying vertices
- Drawing is specified in world coordinates.
- Drawing is performed in screen coordinates.
- A number of coordinate transformations.
- Rasterization – connecting / filling the dots.